

Continuous Wavelet Transform based CNN Model for EV Battery State of Charge Estimation

Prasanta Kumar Mohanty, Premalata Jena, Narayana Prasad Padhy

Department of Electrical Engineering

IIT Roorkee, Roorkee, India

pkumarmohanty@ee.iitr.ac.in, premalata.jena@ee.iitr.ac.in, nppadhy@ee.iitr.ac.in

Abstract— The paper introduces a novel approach called Continuous Wavelet Transform-based Convolutional Neural Network (CWT-CNN) for precisely estimating electric vehicle battery state-of-charge (SOC). SOC holds significant importance in analysing vehicle range and ensuring battery cell balancing. To validate the effectiveness of the proposed model, three real-world datasets are utilised for both the training and testing stages. Incorporating the Continuous Wavelet Transform (CWT) in the proposed CWT-CNN model plays a vital role. CWT enables extracting essential features from the input data by analysing signals in both time and frequency domains. This process aids in capturing significant patterns and variations that might not be easily discernible through conventional methods. To evaluate the accuracy of the CWT-CNN model's predictions on the test set data, two evaluation metrics, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), are employed. The reported highest percentage values of RMSE (1.26%) and MAE (0.76%) demonstrate the effectiveness of the proposed CWT-CNN model in accurately estimating the state-of-charge (SOC) of electric vehicle batteries.

Keywords—Electric vehicle, Deep learning, State-of-charge, Wavelet transform, Convolutional neural network.

I. INTRODUCTION

The necessity of electric vehicle integration in the transportation sector arises from its crucial role in mitigating air pollution. The energy storage system employed in these vehicles is pivotal in determining their cost, performance, size, and safety. In contemporary times, Lithium-ion batteries have gained significant preference in electric vehicles owing to their superior energy efficiency. The performance of a battery is influenced by a multitude of parameters, including battery cell voltage, current, state of charge (SOC), cell temperature, ambient temperature, and more. These factors, in turn, have a direct impact on various aspects, such as the vehicle's range, the cell balancing process, energy consumption, and the overall battery life. In battery management systems, battery SOC (State of Charge) is a key factor. Since the battery's SOC has a significant impact on both its general performance and efficiency, determining an accurate estimate of the SOC is of the utmost importance.

In addition to conventional model-driven approaches, data-driven AI techniques are a key feature of contemporary

literature. These AI-driven techniques are becoming more and more well-liked, and this is largely because of their extraordinary capacity to handle massive amounts of data effectively. These AI-powered, data-driven tools process and analyze massive volumes of information, producing more precise and insightful outcomes. Due to their success in managing complicated and high-dimensional data, they have consequently become a preferred option for numerous applications extending across various sectors.

Furthermore, these methods have proven to be more precise in predicting a wide range of outcomes, thus becoming invaluable in battery SOC estimation. For instance, in [1], the researchers utilized a 1D CNN model for SOC estimation. Additionally, in [2], they employed a deep LSTM model, achieving impressive RMSE and MAE values of 0.0239 and 0.0202, respectively. In [3], a Gated Recurrent Neural Network (GRU) model was employed, and the corresponding RMSE values were found to be 2.15, 1.30, and 1.34 for 0, 25, and 45 degrees, respectively. In [4], authors explored multiple Artificial Neural Network models, such as Support Vector Machines, CNN, and Gated Recurrent Neural Networks. They discovered that the CNN model outperformed others, yielding RMSE and MAE values of 1.74 (± 0.269) and 1.44 (± 0.216), respectively. Furthermore, in [5], a CNN-based model was proposed and optimised using specific parameters for improved performance. However, a crucial aspect overlooked in the aforementioned literature is the significance of extracting intricate temporal and frequency characteristics from input data, particularly when dealing with limited datasets.

This paper introduces a novel CWT-based CNN model, called CWT-CNN, designed for precise electric vehicle battery state of charge (SOC) prediction. The Continuous Wavelet Transform (CWT) is employed to extract complex temporal and frequency characteristics from the input data, and its resulting output is subsequently fed into a CNN model for the SOC prediction task. The paper highlights three main contributions:

- 1) The utilisation of CWT enhances prediction accuracy, even with limited input data, eliminating the need for complex models in SOC prediction.

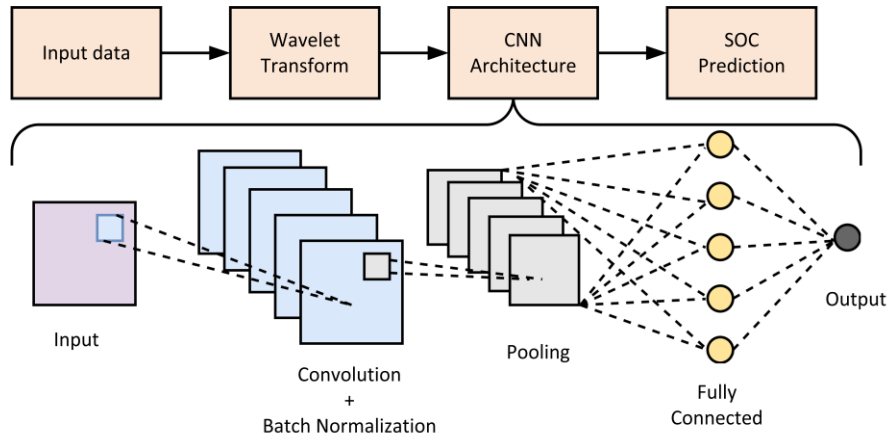


Fig. 1. Architecture of the proposed CWT-CNN model

- 2) The features extracted from CWT serve as input for a CNN-based regression model, enabling precise SOC prediction.
- 3) The overall model is trained and tested on three publicly available datasets: DST, US06, and FUDS, encompassing current and voltage data across three different ambient temperatures (0, 25, and 45 degrees).

The remainder of the paper is structured as follows. Section II provides a detailed explanation of the proposed technique. It delves into the architecture and functioning of the CWT-CNN model, highlighting how it captures important features for SOC prediction. In Section III, the simulation results are presented, showcasing the model's performance on various datasets and scenarios. Finally, Section IV concludes the paper, summarizing the key findings and contributions. It also discusses potential areas for future research and development in the field of battery SOC prediction using advanced data-driven techniques.

II. PROPOSED METHODOLOGY

In this section, three subsections are included, namely the "Proposed CWT-CNN Model," "Input Data," and "Evaluation Metrics." This section provides a comprehensive understanding of the proposed methodology for precise battery SOC prediction, utilizing input parameters such as measured voltage, measured current, and the ambient temperature under which the experiments were conducted.

A. CWT-CNN Model

Fig. 1 illustrates the complete architecture of the proposed CWT-based CWT-CNN model. Section II (B) provides details about the input dataset. The raw dataset is input into the continuous wavelet transform (CWT) block, which generates a scalogram map representing time-frequency representation images. These images are capable of capturing low frequencies with improved frequency resolution and high

frequencies with enhanced time resolution, owing to the variable scale wavelets used [6]. Moreover, the scalogram wavelet transform demonstrates superior analytical capabilities when dealing with signals exhibiting highly

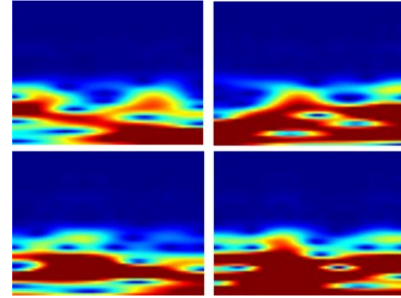


Fig. 2. Four random scalogram images generated from the DST

distorted shapes. Fig. 2 displays four randomly selected scalogram images from the comprehensive input training dataset. These images are generated using the CWT block applied to the dynamic test profile data referred to as DST. These are 2D images having both spatial and temporal information of the input time series 1D datasets.

After generating the scalogram images, they are subsequently fed into the CNN regression model for battery SOC prediction [7]. The CNN model proposed in this study consists of three sets of consecutive four-layer pairs. Each of these pairs comprises the following layers: a convolution layer, a batch normalization layer, a Rectified Linear Unit (ReLU) layer, and a Max pooling layer. This architecture is designed to effectively extract and learn important features from the input data, contributing to accurate predictions in the task of estimating the state-of-charge (SOC) of electric vehicle batteries. These layer combinations are designed to extract and learn more intricate features present in the scalogram images, enhancing the model's ability to make accurate SOC predictions. Following the extraction of features from the previous layers, a dropout layer is incorporated, which helps in regularizing the model and

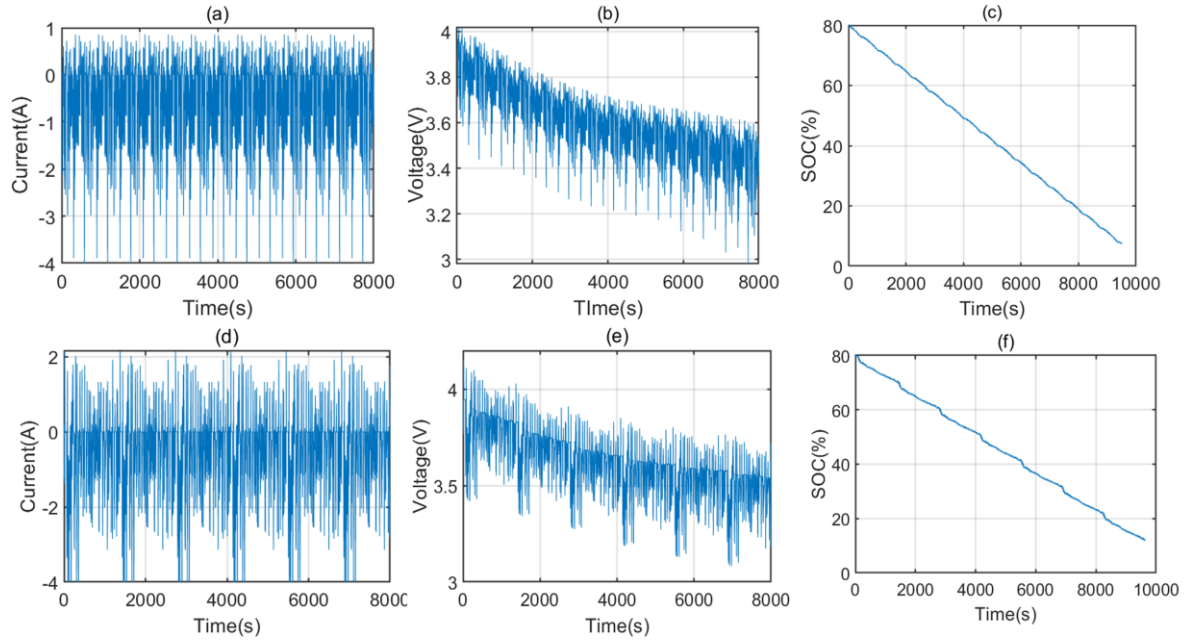


Fig. 3. (a) Measured current profile for US06 profile; (b) measured voltage profile for US06 profile; (c) SOC values for US06 profile; (d) measured current profile for FUDS profile; (e) measured voltage profile for FUDS profile; (f) SOC values for FUDS profile.

preventing overfitting. After the dropout layer, a fully connected layer is employed to combine the learned features. Finally, a regression layer serves as the output layer to produce the predicted battery SOC values.

The optimization task in the proposed model utilizes the SGDM (Stochastic Gradient Descent with Momentum) optimizer. During training, a minibatch size of 64 is employed, and the training process is limited to a maximum of 30 epochs. The training progress is monitored and evaluated based on two metrics: the training Root Mean Square Error (RMSE) and the Loss function.

B. Input Dataset

The input dataset utilized in this paper comprises publicly available datasets provided by the CALCE battery research team [8]. The Li-ion battery model specifications are detailed in Table I. The dynamic test profile data consists of measured voltage and measured current profiles recorded at three distinct temperatures (0, 25, and 45 degrees Celsius). These profiles are obtained using three driving cycles named DST, US06, and FUDS. The actual SOC profiles are generated and recorded using those voltage, current, and ambient

temperature profiles. Fig. 3 (a-c) shows the current, voltage, and actual SOC profile for the US06 dataset. Similarly, Fig. 3 (d-f) shows the current, voltage, and actual SOC profile for the FUDS dataset. The training data for the model consists of the current and voltage profiles from the DST data at 0 degrees Celsius. On the other hand, the parameters from the US06 and FUDS data are used for separate testing of the model. It is also to be noted that the highest value of the battery SOC is set to 80%.

C. Evaluation Metrics

The performance of the proposed CWT-CNN model is assessed using two evaluation metrics: RMSE and MAE. Moreover, during the training process, RMSE values and loss functions are computed for each minibatch. The training loss function is defined as half of the mean square error between the predicted SOC values (represented by $\hat{y}_m^{predicted}$) and the true or targeted SOC values (represented by y_m^{actual}) for all responses. The expressions for the above metrics are expressed by:

$$Loss Function = \frac{1}{2} \sum_{m=1}^M (y_m^{actual} - \hat{y}_m^{predicted})^2 \quad (1)$$

$$RMSE = \sqrt{\frac{1}{M} \sum_{m=1}^M (y_m^{actual} - \hat{y}_m^{predicted})^2} \quad (2)$$

$$MAE = \frac{1}{M} \sum_{m=1}^M |y_m^{actual} - \hat{y}_m^{predicted}| \quad (3)$$

TABLE I
BATTERY SPECIFICATIONS

Parameters	Specifications
Model	INR 18650-20R
Capacity	2000 mAh
Cell Chemistry	LiNiMnCo/Graphite
Weight	45g

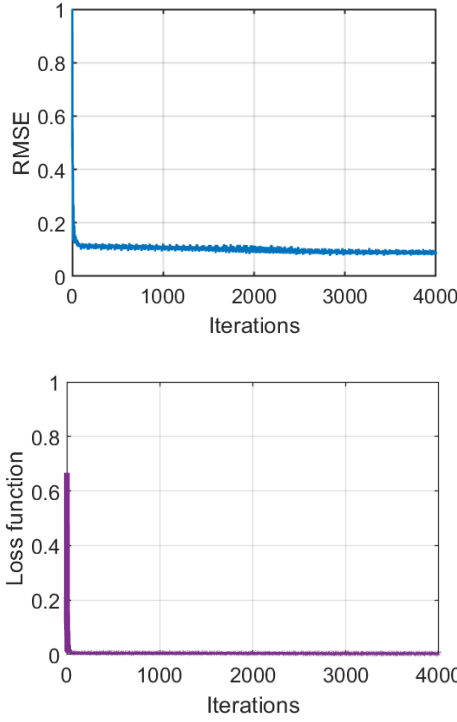


Fig. 4. Training RMSE values and loss function variation over 30 epochs

III. SIMULATION RESULTS

This section presents the validation of the proposed CWT-CNN model through simulation results. The validation process begins by examining the training outcomes, followed by the analysis of the final SOC comparison graphs. This comprehensive evaluation ensures the model's effectiveness and accuracy in predicting battery SOC. The training data used in this study consists of the measured current and measured voltages recorded at an ambient temperature of 0 degrees Celsius, employing the DST driving cycles. On the other hand, the testing dataset comprises SOC values from approximately 10,000 samples taken from three datasets: DST, US06, and FUDS, all obtained at an ambient temperature of 25 degrees Celsius. This division enables the model to learn from the training data and subsequently evaluate its performance on unseen data from different driving cycles and temperature conditions. In Fig. 4, the training Root Mean Square Error (RMSE) and loss function values for the DST dataset are depicted. It is evident that both values consistently decrease and reach levels well below 0.1

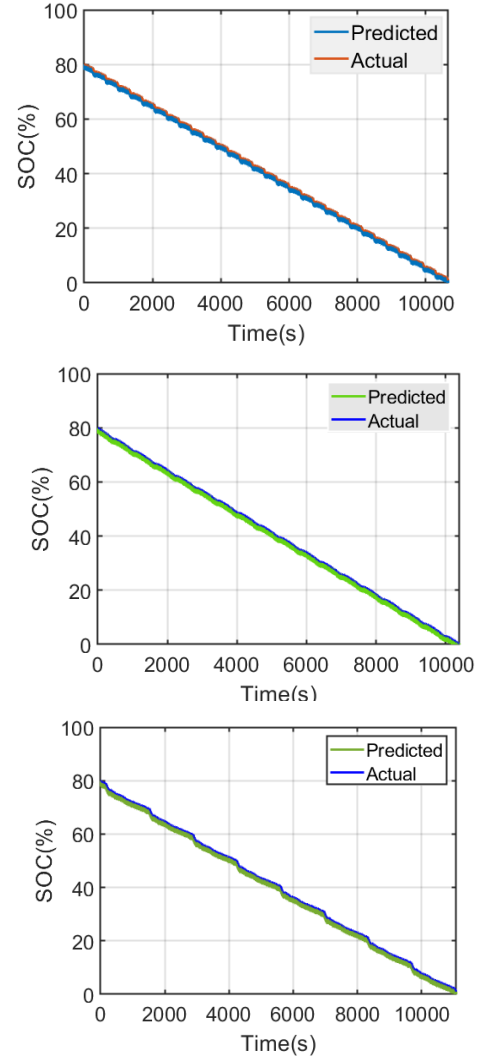


Fig. 5. Percentage values of the battery SOC comparisons for the DST, US06, and FUDS datasets at 25 degrees Celsius.

after the 100th iteration. This significant reduction in RMSE and loss function values demonstrates the effectiveness of the proposed model in accurately predicting the battery SOC for the DST dataset.

After completing the training process, the model is subjected to testing using three previously unseen datasets to predict the battery SOC values. The SOC comparison results are displayed in Fig. 5. It is evident from Fig. 5 that the predicted SOC values closely replicate the actual values for all three different datasets. This indicates the robustness and accuracy of the proposed model in generalizing to new data and making reliable predictions across different driving cycles and ambient temperatures. Additionally, to assess the model's performance, evaluation metrics such as RMSE and MAE percentages are computed and presented in Table II. Remarkably, the highest values of RMSE and MAE percentages are determined to be 1.26% and 0.76%, respectively. These values are considerably lower when compared to existing techniques, indicating the superior

TABLE II
TEST SET EVALUATION METRICS VALUES

Dataset	RMSE (%)	MAE (%)
DST	1.10	0.60
US06	1.05	0.52
FUDS	1.26	0.76

predictive capabilities of the proposed CWT-CNN model. The model's ability to achieve such low percentage values demonstrates its effectiveness in accurately estimating battery SOC.

IV. CONCLUSION AND FUTURE SCOPE

This paper introduces a novel and advanced deep learning technique called the CWT-CNN model. Unlike traditional CNN models, the CWT-CNN model incorporates a dedicated Continuous Wavelet Transform (CWT)-based feature extraction block before the CNN network. This unique addition enables the model to effectively capture essential spatial and temporal features from the complex input data, enhancing its accuracy and efficiency in predicting the state of charge (SOC) of electric vehicle batteries. Indeed, the maximum percentage values of RMSE (1.26%) and MAE (0.76%) achieved in the evaluation of the proposed model provide strong evidence of its effectiveness. These low percentage values indicate that the model's predictions are highly accurate and closely aligned with the actual values of the battery's state of charge (SOC). The future prospects of this paper encompass two main dimensions. Firstly, there is potential for investigating the influence of battery aging, degradation, and external factors (such as weather conditions and driving behavior) on battery SOC. Integrating these factors into the prediction models would enhance the model's practicality and accuracy. Secondly, employing transfer learning to reduce computational time during long-term SOC prediction scenarios could significantly optimize battery usage and management. By adopting transfer learning techniques, the model can efficiently adapt and improve its performance for long-term forecasting tasks, making it more practical and applicable in real-world applications.

REFERENCES

- [1] A. Bhattacharjee, A. Verma, S. Mishra, and T. K. Saha, "Estimating State of Charge for xEV Batteries Using 1D Convolutional Neural Networks and Transfer Learning," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3123–3135, 2021, doi: 10.1109/TVT.2021.3064287.
- [2] P. K. Mohanty, P. Jena, and N. P. Padhy, "Electric Vehicle State-of-charge Prediction using Deep LSTM Network Model," *10th IEEE Int. Conf. Power Electron. Drives Energy Syst. PEDES 2022*, pp. 1–6, 2022, doi: 10.1109/PEDES56012.2022.10080466.
- [3] Z. Zhang *et al.*, "An Improved Bidirectional Gated Recurrent Unit Method for Accurate State-of-Charge Estimation," *IEEE Access*, vol. 9, pp. 11252–11263, 2021, doi: 10.1109/ACCESS.2021.3049944.
- [4] S. Pohlmann, D. Karnehm, A. Mashayekh, M. Kuder, A. Gieraths, and T. Weyh, "State-of-Charge Estimation of Lithium-Ion Batteries Using Machine Learning Based on Augmented Data," *SEST 2022 - 5th Int. Conf. Smart Energy Syst. Technol.*, pp. 1–6, 2022, doi: 10.1109/SEST53650.2022.9898474.
- [5] D. Li and L. Yang, "Remaining useful life prediction of lithium battery using convolutional neural network with optimized parameters," *Proc. - 2020 5th Asia Conf. Power Electr. Eng. ACPEE 2020*, no. 51875339, pp. 840–844, 2020, doi: 10.1109/ACPEE48638.2020.9136289.
- [6] T. Guo, T. Zhang, E. Lim, M. Lopez-Benitez, F. Ma, and L. Yu, "A Review of Wavelet Analysis and Its Applications: Challenges and Opportunities," *IEEE Access*, vol. 10, pp. 58869–58903, 2022, doi: 10.1109/ACCESS.2022.3179517.
- [7] B. Y. Adhaityar, D. P. Sahara, C. Pratama, A. Wibowo, and L. S. Heliani, "Multi-Target Regression Using Convolutional Neural Network-Random Forests (CNN-RF) for Early Earthquake Warning System," *2021 9th Int. Conf. Inf. Commun. Technol. ICoICT 2021*, pp. 31–36, 2021, doi: 10.1109/ICoICT52021.2021.9527461.
- [8] University of Maryland, "Center for Advanced Life Cycle Engineering," *Sch. Eng.*, no. 301, pp. 1–20, 2019, Accessed: Jul. 28, 2023. [Online]. Available: <https://calce.umd.edu/>